

AN INTRODUCTION TO REGRESSION TECHNIQUES

20th August 2026

GOING BACK TO THE START

Regression relates the expected values of one random variable, Y , to the known values of associated variable X . The term regression comes from Sir Francis Galton, a geneticist who studied the sizes of seeds, their offspring, and the heights of fathers and sons. In both cases, he found that the offspring of larger than average size parents tended to be smaller than their parents and that the offspring of smaller than average size parents tended to be larger than their parents. He called this regression towards mediocrity. Note that the variables, X , which are the independent variables, are considered as fixed values in regression. They are also known as explanatory or predictor variables. The variable, Y , is the dependent variable.

SIMPLE LINEAR REGRESSION

The aim of a linear regression model is to fit a line that predicts the value of the y variable as best possible given x . We should expect a line, $y = \beta_1 + \beta_2 x$, but in reality the linear fit is not exact, there are errors, and thus, the model is written as $y_i = \beta_1 + \beta_2 x_i + e_i$, where β_1 is the intercept parameter, β_2 is the slope parameter $\left(\frac{dy}{dx}\right)$, and e_i represents the error in the linear fit for the i^{th} observation. Among all possible straight lines, to obtain a line of best fit we find the values of β_1 and β_2 that minimise $S(\beta_1, \beta_2) = \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2$. These values of β_1 and β_2 correspond to the least squares estimates.

By the Gauss-Markov theorem, for the least squares estimator b_2 of β_2 has the smallest variance among the class of linear and unbiased estimators, *i.e.*, it is the best linear unbiased estimator, BLUE. For example, to be unbiased, $\mathbb{E}[b_2] = \beta_2$, and to be linear, we can rewrite β_2 as $\sum w_i y_i$, where $w_i = \frac{(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$, a constant.

2.1 Deriving the Least Squares Estimates

To obtain the least squares estimates, we find the partial derivatives of S with respect to β_1 and β_2 . Let $\sum_{i=1}^n = \sum$ for ease of notation. Thus:

$$\begin{aligned}\frac{\partial S}{\partial \beta_1} &= 2n\beta_1 - 2 \sum y_i + 2 \left(\sum x_i \right) \beta_2, \\ \frac{\partial S}{\partial \beta_2} &= 2 \left(\sum x_i^2 \right) \beta_2 - 2 \sum x_i y_i + 2 \left(\sum x_i \right) \beta_1.\end{aligned}$$

Set the partial derivatives equal to 0, and replace β_1 and β_2 by $b_1 = \hat{\beta}_1$ and $b_2 = \hat{\beta}_2$, respectively.

$$\begin{aligned}2 \left[\sum y_i - nb_1 - \left(\sum x_i \right) b_2 \right] &= 0, \\ 2 \left[\sum x_i y_i - \left(\sum x_i \right) b_1 - \left(\sum x_i^2 \right) b_2 \right] &= 0.\end{aligned}$$

After simplification, we have the normal equations:

$$nb_1 + \left(\sum x_i \right) b_2 = \sum y_i, \quad (1)$$

$$\left(\sum x_i \right) b_1 + \left(\sum x_i^2 \right) b_2 = \sum x_i y_i \quad (2)$$

We now solve the two equations simultaneously, for b_2 , multiply (1) by $\sum x_i$, and multiply (2) by n , then subtract the first equation from the second, whilst isolating b_2 on the LHS.

$$b_2 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - \left(\sum x_i \right)^2}. \quad (3)$$

To solve for b_1 , given b_2 , divide both sides of (1) by n and rearrange such that $b_1 = \bar{y} - b_2\bar{x}$. We can rewrite (3) in mean form as $b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{XY}}{S_{XX}}$, where $S_{XY} = \sum (x_i - \bar{x})(y_i - \bar{y})$, $S_{XX} = \sum (x_i - \bar{x})^2$.

Thus, the line of best fit is $y - \bar{y} = \frac{S_{XY}}{S_{XX}}(x - \bar{x})$, i.e., the line passes through $(\bar{x}, \bar{y})$. We denote the fitted values of y by $\hat{y}$, thus, $\hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 x_i = \bar{y} + \frac{S_{XY}}{S_{XX}}(x_i - \bar{x})$, where the values $e_i = y_i - \hat{y}_i$ denote the residuals of the fitted model. Therefore, the sample correction coefficient can be written as:

$$r_{XY} = \frac{\frac{1}{n-1} (\sum (x_i - \bar{x})(y_i - \bar{y}))}{\sqrt{\frac{1}{n-1} (\sum (x_i - \bar{x})^2) \frac{1}{n-1} (\sum (y_i - \bar{y})^2)}} = \frac{S_{XY}}{\sqrt{S_{XX} S_{YY}}}.$$

2.2 Assumptions of a Linear Regression Model

1. The value of y , for each value of x , is $y_i = \beta_1 + \beta_2 x_i + e_i$.
2. The expected value of the random error, e , is $\mathbb{E}[e] = 0 \Leftrightarrow \mathbb{E}[y] = \beta_1 + \beta_2 x$.
3. The variance of the random error, e , and y is constant: $Var(e_i, e_j) = \sigma^2$. The random variables, y and e , have the same variance because they differ only by a constant. This is called homoskedasticity (or homoscedasticity).
4. The covariance between any pair of random errors, e_i and e_j , is: $Cov(e_i, e_j) = Cov(y_i, y_j) = 0$. The stronger version of this assumption is that the random errors, e , are statistically independent, in which case, the values of the dependent variable, y , are also statistically independent.
5. The variable, x , is not random, and must take at least two different values.
6. (Optional) The values of e are normally distributed about their mean if the values of y are normally distributed, and vice versa, $y \sim N(\beta_1 + \beta_2 x, \sigma^2) \Leftrightarrow e \sim N(0, \sigma^2)$.

Do note that, in practice, some of these assumptions are violated and the simple linear least squares regression is no longer the BLUE.

MULTIPLE LINEAR REGRESSION

Why do we need to study models with more than one predictor/independent variable? By doing so, it can help us investigate more information for descriptive inference by accounting for more factors, and thus, improve the fit and predictive power of the model. It also allows us to control for factors for causal inference such as non-linearities and interactive effects. In a general multiple regression model, a dependent variable, y , is related to a number of explanatory variables, $x_{2i}, x_{3i}, \dots, x_{Ki}$ through a linear equation that can be written as $y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \beta_K x_{Ki} + e_i$, where i denotes the individual observations $i = 1, 2, \dots, n$. A single parameter, β_k , measures the effect of a change in the variable x_k upon the expected value of y , all other variables held constant, $\beta_k = \frac{\Delta \mathbb{E}[y]}{\Delta x_k} \Big|_{\text{other } x\text{'s held constant}} = \frac{\partial \mathbb{E}[y]}{\partial x_k}$. As same as linear regression, the parameter, β_1 , is the intercept term, of which we can think of it being attached to a variable, x_{1i} , that is always equal to 1.

To find the least squares estimates, we utilise the same method as above, mathematically minimise the sum of the squares function $S(\beta_1, \beta_2, \dots, \beta_K)$. However, as the number of predictors increases, the resulting system of equations becomes increasingly complex and difficult to solve by hand and are most conveniently expressed using matrix algebra. Consequently, the estimates can be obtained using statistical software, such as the `lm()` function in R, rather than through manual calculation, which we will expand on in future articles.

The Gauss-Markov theorem is not limited to simple linear regression and extends to multiple linear regression. Thus, the least squares estimators are the BLUE.

3.1 Assumptions of a Multiple Regression Model

The assumptions for multiple linear regression are essentially the same as those for simple linear regression, but they are extended to account for all K explanatory variables rather than a single predictor.

1. $y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \beta_K x_{Ki} + e_i$, $i = 1, 2, \dots, n$.
2. $\mathbb{E}[e_i] = 0 \Leftrightarrow \mathbb{E}[y_i] = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \beta_K x_{Ki}$.
3. $Var(y_i) = Var(e_i) = \sigma^2$.
4. $Cov(e_i, e_j) = Cov(y_i, y_j) = 0$.
5. The values of each x_{ki} are not random and not exact linear functions of the other explanatory variables.
6. $y_i \sim N(\beta_1 + \beta_2 x_{2i} + \dots + \beta_K x_{Ki}, \sigma^2) \Leftrightarrow e \sim N(0, \sigma^2)$.

CONCLUSION

We have introduced both simple and multiple linear regression models, and listed the assumptions that underpin their use. These models form the foundation of regression analysis. Regression can be applied throughout finance and economics, and to analysts, regression techniques are indispensable.

ALEXANDER FLANN